

ANALYSIS 1

PREPARATION FOR TEST 1

Exercise 1. *Let*

$$f(x) := \sqrt{x+3} \quad (-3 \leq x \leq 6) \quad \text{and} \quad g(x) := x^2 + 4x + 1 \quad (x \leq -3).$$

- a) *Determine the composite function $f \circ g$.*
b) *Prove that the function g is invertible, and find its inverse.*

Solution

a) $\mathcal{D}_{f \circ g} = \{x \in \mathcal{D}_g \mid g(x) \in \mathcal{D}_f\} = \{x \leq -3 \mid -3 \leq x^2 + 4x + 1 \leq 6\}.$

$$x^2 + 4x + 1 \geq -3 \iff x^2 + 4x + 4 = (x+2)^2 \geq 0$$

which is true for all $x \in \mathbb{R}$.

$$x^2 + 4x + 1 \leq 6 \iff x^2 + 4x - 5 = (x+5)(x-1) \leq 0 \iff x \in [-5, 1].$$

Since $x \leq -3$, we have $\mathcal{D}_{f \circ g} = [-5, -3]$. Then

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) = \sqrt{(x^2 + 4x + 1) + 3} = \sqrt{x^2 + 4x + 4} = \sqrt{(x+2)^2} = |x+2| \\ &= -x-2 \quad (-5 \leq x \leq -3). \end{aligned}$$

b) First, note that

$$g(x) = x^2 + 4x + 1 = (x+2)^2 - 3 \quad (x \leq -3).$$

Let $x, t \in \mathcal{D}_g = [-\infty, -3]$. Then

$$\begin{aligned} g(x) = g(t) &\implies (x+2)^2 - 3 = (t+2)^2 - 3 \implies (x+2)^2 = (t+2)^2 \\ &\implies |x+2| = |t+2| \xrightarrow{x+2 < 0, t+2 < 0} -(x+2) = -(t+2) \\ &\implies x+2 = t+2 \implies x = t. \end{aligned}$$

Thus, g is invertible.

To obtain g^{-1} we determine first $\mathcal{D}_{g^{-1}} = \mathcal{R}_g$. Since $\mathcal{D}_g = (-\infty, -3]$, then if $x \in \mathcal{D}_g$ we have

$$x \leq -3 \iff x+2 \leq -1 \iff -(x+2) \geq 1 \xrightarrow{x+2 < 0} |x+2| \geq 1$$

$$\iff (x+2)^2 \geq 1 \iff g(x) = (x+2)^2 - 3 \geq -2.$$

Thus, $\mathcal{D}_{g^{-1}} = \mathcal{R}_g = [-2, +\infty)$.

On the other hand, if $x \in \mathcal{D}_g$, so $x \leq -3$, and $y \in \mathcal{D}_{g^{-1}}$, so $y \geq -2$, we have

$$\begin{aligned} y = g(x) &\iff y = (x+2)^2 - 3 \iff y+3 = (x+2)^2 \xLeftrightarrow[y+3>0] \sqrt{y+3} = |x+2| \\ &\xLeftrightarrow[x+2<0] \sqrt{y+3} = -x-2 \iff x = -2 - \sqrt{y+3}. \end{aligned}$$

Thus, $g^{-1}(y) = -2 - \sqrt{y+3} \quad (y \geq -2)$.

Exercise 2. Determine (without using the concept of the limit) $\sup H$, $\inf H$, $\max H$ and $\min H$ if

$$H = \left\{ \frac{3|x| - 1}{4|x| + 2} \in \mathbb{R} \mid x \in [-3, +\infty) \right\}.$$

Solution The set

$$H = \left\{ \frac{3|x| - 1}{4|x| + 2} \in \mathbb{R} \mid x \in [-3, +\infty) \right\}$$

is nothing more than the range of the function

$$h(x) := \frac{3|x| - 1}{4|x| + 2} \quad (x \in [-3, +\infty)).$$

Notice that

$$\left\{ \frac{3|x| - 1}{4|x| + 2} \in \mathbb{R} \mid x \in [-3, +\infty) \right\} = \left\{ \frac{3|x| - 1}{4|x| + 2} \in \mathbb{R} \mid x \in [-3, 0) \right\} \cup \left\{ \frac{3|x| - 1}{4|x| + 2} \in \mathbb{R} \mid x \in [0, +\infty) \right\},$$

and

$$h(-x) = \frac{3|-x| - 1}{4|-x| + 2} = \frac{3|x| - 1}{4|x| + 2} = h(x) \quad (-3 \leq x \leq 3).$$

Therefore,

$$\left\{ \frac{3|x| - 1}{4|x| + 2} \in \mathbb{R} \mid x \in [-3, 0) \right\} \subset \left\{ \frac{3|x| - 1}{4|x| + 2} \in \mathbb{R} \mid x \in [0, +\infty) \right\} = \left\{ \frac{3x - 1}{4x + 2} \in \mathbb{R} \mid x \in [0, +\infty) \right\},$$

from which it follows that

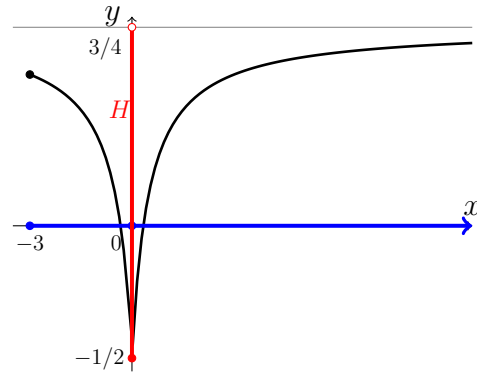
$$H = \left\{ \frac{3x - 1}{4x + 2} \in \mathbb{R} \mid x \in [0, +\infty) \right\}.$$

Let us rewrite the function h :

$$h(x) = \frac{3x - 1}{4x + 2} = \frac{\frac{3}{4}(4x + 2) - \frac{3}{2} - 1}{4x + 2} = \frac{\frac{3}{4}(4x + 2) - \frac{5}{2}}{4x + 2} = \frac{3}{4} - \frac{5}{8x + 4}.$$

It is evident that $h(x) < 3/4$, since $5/(8x+4) > 0$ for $x \geq 0$. Moreover, h is monotonically increasing on the set $[0, +\infty)$, as

$$x \uparrow \implies 8x + 4 \uparrow \implies \frac{1}{8x + 4} \downarrow \implies -\frac{5}{8x + 4} \uparrow \implies \frac{3}{4} - \frac{5}{8x + 4} \uparrow.$$



Due to the monotonic increase, $h(x) \geq h(0) = \frac{0-1}{0+2} = -\frac{1}{2}$ ($x \geq 0$), and $-\frac{1}{2} = h(0) \in H$. Therefore,

$$\min H = \inf H = -\frac{1}{2}.$$

Conjecture: $\sup H = \frac{3}{4}$.

Proof.

- i) We have already seen that $h(x) < 3/4$ ($x \geq 0$), so $3/4$ is an upper bound of H .
- ii) We will show that $\frac{3}{4}$ is the least upper bound of H . This means that any number less than $\frac{3}{4}$ is not an upper bound of H , that is,

$$\begin{aligned} & \forall \varepsilon > 0 \exists h \in H : h > \frac{3}{4} - \varepsilon \\ (\#) \quad & \iff \forall \varepsilon > 0 \exists x \geq 0 : h(x) = \frac{3}{4} - \frac{5}{8x+4} > \frac{3}{4} - \varepsilon \\ & \iff \forall \varepsilon > 0 \exists x \geq 0 : \frac{5}{8x+4} < \varepsilon. \end{aligned}$$

Let us take an arbitrarily fixed number $\varepsilon > 0$. Observe that if $x > 0$, then

$$\frac{5}{8x+4} < \frac{5}{8x} < \frac{1}{x} \quad \text{and} \quad \frac{1}{x} < \varepsilon \iff x > \frac{1}{\varepsilon}.$$

Then, by choosing $x = \frac{1}{\varepsilon} + 1$, $(\#)$ holds.

Since both i) and ii) hold, we conclude that $\sup H = \frac{3}{4}$.

However, there does not exist any $x \geq 0$ such that $h(x) = \frac{3}{4} - \frac{5}{8x+4} = \frac{3}{4}$, since $\frac{5}{8x+4} \neq 0$. Therefore, $\nexists \min H$.

Exercise 3. Evaluate the following limit, and prove by definition your result.

$$\lim_{n \rightarrow +\infty} \frac{3n^2 + 2n - 10}{6n^2 - 15n + 1}.$$

Solution Using the rules for calculating limits

$$\lim_{n \rightarrow +\infty} \frac{3n^2 + 2n - 10}{6n^2 - 15n + 1} = \lim_{n \rightarrow +\infty} \frac{\cancel{n^2}}{\cancel{n^2}} \cdot \frac{3 + \frac{2}{n} - \frac{10}{n^2}}{6 - \frac{15}{n} + \frac{1}{n^2}} = \frac{3 + 0 - 0}{6 - 0 + 0} = \frac{1}{2}.$$

According to the definition of limit, we need to prove that

$$(*) \quad \forall \varepsilon > 0 \exists n_0 \in \mathbb{N}, \forall n > n_0 : \left| \frac{3n^2 + 2n - 10}{6n^2 - 15n + 1} - \frac{1}{2} \right| < \varepsilon.$$

We will estimate the following expression from above

$$\left| \frac{3n^2 + 2n - 10}{6n^2 - 15n + 1} - \frac{1}{2} \right| = \left| \frac{2(3n^2 + 2n - 10) - (6n^2 - 15n + 1)}{2(6n^2 - 15n + 1)} \right| = \frac{1}{2} \cdot \frac{|19n - 21|}{|6n^2 - 15n + 1|}.$$

Estimating the numerator from above and the denominator from below:

$$|19n - 21| \leq |19n| + |-21| = 19n + 21 \underset{(n \geq 1)}{\leq} 19n + 21n = 40n$$

and

$$6n^2 - 15n + 1 \geq 6n^2 - 15n = n^2 + 5n(n-3) \underset{(n \geq 3)}{\geq} n^2 + 0 = n^2 > 0.$$

Hence, if $n \geq 3$, then $6n^2 - 15n + 1 > 0 \implies |6n^2 - 15n + 1| = 6n^2 - 15n + 1 \geq n^2$, we have

$$\frac{1}{2} \cdot \frac{|19n - 21|}{|6n^2 - 15n + 1|} \leq \frac{1}{2} \cdot \frac{40n}{n^2} = \frac{20}{n} = \underbrace{\frac{20}{n}}_{n > \frac{20}{\varepsilon}} < \varepsilon.$$

Therefore $(*)$ holds for all

$$n > n_0 := \left\lceil \frac{20}{\varepsilon} \right\rceil + 3.$$

Exercise 4. Determine the following limits

$$a) \quad \lim_{n \rightarrow +\infty} \left(\sqrt{n^4 + n + 3} - \sqrt{n^4 - n^2 + 1} \right),$$

$$b) \quad \lim_{n \rightarrow +\infty} \sqrt[n]{\frac{n+1}{2^n + n^2}},$$

$$c) \quad \lim_{n \rightarrow +\infty} \left(\frac{3n+1}{3n-2} \right)^{2n+3}.$$

Solution

1. By multiplying and dividing by its conjugate:

$$\begin{aligned} \sqrt{n^4 + n + 3} - \sqrt{n^4 - n^2 + 1} &= \frac{(n^4 + n + 3) - (n^4 - n^2 + 1)}{\sqrt{n^4 + n + 3} + \sqrt{n^4 - n^2 + 1}} \\ &= \frac{n^2 + n + 2}{\sqrt{n^4 + n + 3} + \sqrt{n^4 - n^2 + 1}} = \frac{n^2 + n + 2}{n^2 \sqrt{1 + \frac{1}{n^3} + \frac{3}{n^4}} + n^2 \sqrt{1 - \frac{1}{n^2} + \frac{1}{n^4}}} \\ &= \frac{1 + \frac{1}{n} + \frac{2}{n^2}}{\sqrt{1 + \frac{1}{n^3} + \frac{3}{n^4}} + \sqrt{1 - \frac{1}{n^2} + \frac{1}{n^4}}} \longrightarrow \frac{1 + 0 + 0}{\sqrt{1 + 0 + 0} + \sqrt{1 - 0 + 0}} = \frac{1}{2}. \end{aligned}$$

2.

$$\sqrt[n]{\frac{n+1}{2^n + n^2}} = \sqrt[n]{\frac{n}{2^n} \cdot \frac{1 + \frac{1}{n}}{1 + \frac{n^2}{2^n}}} = \frac{\sqrt[n]{n}}{2} \sqrt[n]{\frac{1 + \frac{1}{n}}{1 + n^2(\frac{1}{2})^n}} \longrightarrow \frac{1}{2} \cdot 1 = \frac{1}{2},$$

since

$$x_n := \frac{1 + \frac{1}{n}}{1 + n^2(\frac{1}{2})^n} \longrightarrow \frac{1 + 0}{1 + 0} = 1 \in (0, +\infty) \implies \sqrt[n]{x_n} \longrightarrow 1.$$

3.

$$\begin{aligned} \left(\frac{3n+1}{3n-2}\right)^{2n+3} &= \left[\frac{3n \cdot \left(1 + \frac{1}{3n}\right)}{3n \cdot \left(1 - \frac{2}{3n}\right)}\right]^{2n+3} = \left[\frac{\left(1 + \frac{1/3}{n}\right)}{\left(1 + \frac{-2/3}{n}\right)}\right]^{2n+3} = \frac{\left(1 + \frac{1/3}{n}\right)^{2n+3}}{\left(1 + \frac{-2/3}{n}\right)^{2n+3}} \\ &= \frac{\left[\left(1 + \frac{1/3}{n}\right)^n\right]^2 \cdot \left(1 + \frac{1/3}{n}\right)^3}{\left[\left(1 + \frac{-2/3}{n}\right)^n\right]^2 \cdot \left(1 + \frac{-2/3}{n}\right)^3} \xrightarrow{n \rightarrow +\infty} \frac{\left(e^{1/3}\right)^2 \cdot 1^3}{\left(e^{-2/3}\right)^2 \cdot 1^3} = e^{2/3+4/3} = e^2. \end{aligned}$$

Exercise 5. Determine the limits of the following recursive sequence if it exists

$$a_0 = 2, \quad a_{n+1} = \sqrt{5a_n - 4} \quad (n \in \mathbb{N}).$$

Solution The sequence can be written as

$$a_0 = 2, \quad a_{n+1} = f(a_n) \quad (n \in \mathbb{N}).$$

where $f(x) := \sqrt{5x - 4}$ ($x \geq 4/5$). Notice that $f \nearrow$.

We prove by induction that (a_n) is a well-defined sequence, more precisely, we prove that $\forall n \in \mathbb{N}$:

$$(WD) \quad \exists a_n \geq 1.$$

holds.

- (WD) is true for $n = 0$, since we already know that $\exists a_0 = 2 \geq 1$. ✓
- Suppose (WD) is true for some $n \in \mathbb{N}$. Then $5a_n - 4 \geq 1 > 0$, and hence

$$\exists a_{n+1} = \sqrt{5a_n - 4} \geq \sqrt{1} = 1.$$

Therefore (WD) is true for $n + 1$. ✓

In conclusion, we proved by induction that (WD) is true for all $n \in \mathbb{N}$, so (a_n) is a well-defined sequence.

Suppose that $\exists \lim(a_n) =: A \in \mathbb{R}$. By recursion formula

$$A \leftarrow a_{n+1} = \sqrt{5a_n - 4} \rightarrow \sqrt{5A - 4} \quad (n \rightarrow +\infty),$$

since (a_{n+1}) is a subsequence of (a_n) , so it also tends to A . Then, the uniqueness of the limit implies that

$$A = \sqrt{5A - 4},$$

so

$$A^2 = 5A - 4 \implies A^2 - 5A + 4 = 0 \implies (A - 1)(A - 4) = 0 \implies A = 1 \text{ or } A = 4,$$

Thus, if (a_n) is convergent, then its limit is $A = 1$ or $A = 4$.

To study the monotonicity of the sequence (a_n) , notice that

$$a_1 = \sqrt{5a_0 - 4} = \sqrt{5 \cdot 2 - 4} = \sqrt{6} > 2 = a_0.$$

For this reason, we prove by induction that $(a_n) \nearrow$, that is, $\forall n \in \mathbb{N}$:

$$(M) \quad a_n \leq a_{n+1}$$

holds.

- (M) is true for $n = 0$, since we already know that $a_0 \leq a_1$. ✓
- Suppose (M) is true for some $n \in \mathbb{N}$. Then

$$a_{n+1} = \sqrt{5a_n - 4} \leq \left((M) \text{ holds, } f \nearrow \right) \leq \sqrt{5a_{n+1} - 4} = a_{(n+1)+1}$$

Therefore (M) is true for $n + 1$. ✓

In conclusion, we proved by induction that (M) is true for all $n \in \mathbb{N}$, and hence $(a_n) \nearrow$. For this reason, the only possible limit now is $A = 4$.

Finally, we prove by induction that (a_n) is bounded above and $A = 4$ is an upper bound of the sequence, that is, $\forall n \in \mathbb{N}$:

$$(B) \quad a_n \leq 4.$$

holds.

- (B) is true for $n = 0$, since $a_0 = 2 \leq 4$. ✓
- Suppose (B) is true for some $n \in \mathbb{N}$. Then

$$a_{n+1} = \sqrt{5a_n - 4} \leq \left((B) \text{ holds, } f \nearrow \right) \leq \sqrt{5 \cdot 4 - 4} = 4.$$

Therefore (B) is true for $n + 1$. ✓

In conclusion, we proved by induction that (B) is true for all $n \in \mathbb{N}$. Therefore 4 is an upper bound for the sequence (a_n) .

In conclusion (a_n) is a monoton and bounded sequence, so it is convergent. Since only the limit $A = 4$ is possible, thus

$$\lim_{n \rightarrow +\infty} a_n = 4.$$